

How to Install Apache AGE for This Course

Docker is the recommended (and easiest) method.

Apache AGE is a PostgreSQL extension that adds graph database capabilities (openCypher query language). The Docker image ships with PostgreSQL 16 and the AGE 1.6.0 extension pre-installed.

1. Method 1 — Install Apache AGE via Docker (recommended)

This is the cleanest option: no system-level compilation required, easy to reset, and identical across machines.

Step 1 — Install Docker Desktop

1. Download Docker Desktop for your operating system.
2. Install and open it.
3. Make sure Docker is running.

Step 2 — Start Apache AGE using Docker

Make sure you have the `docker-compose.yml` file from this folder available. Ideally, clone the entire repository.

Run this command from the directory containing `docker-compose.yml`:

```
docker compose -f docker-compose.yml up -d
```

Or use the provided script:

```
bash run.sh
```

Note: This container maps port `5433` on your machine to port `5432` inside the container. This avoids conflicts if you also have the PostgreSQL container running on `5432`.

Step 3 — Verify

Run:

```
docker ps
```

You should see a container named `apache_age`. You can also verify with Docker Desktop's GUI.

Connection details:

Parameter	Value
Host	localhost
Port	5433
Username	postgres
Password	postgres
Database	demo

2. Method 2 — Build Apache AGE from Source (advanced)

Only recommended if you need to integrate AGE into an existing PostgreSQL 16 installation.

Step 1 — Prerequisites

- PostgreSQL 16 installed with development headers (`postgresql-server-dev-16` on Debian/Ubuntu).
- Build tools: `git`, `make`, `gcc`.

Step 2 — Clone and build

```
git clone https://github.com/apache/age.git
cd age
git checkout release_PG16_1.6.0
make
sudo make install
```

Step 3 — Enable the extension

Connect to your database and run:

```
CREATE EXTENSION age;
```

3. Install the VS Code PostgreSQL Extension

Apache AGE is built on PostgreSQL, so the same VS Code extension is used.

Step 1 — Open VS Code

Step 2 — Install the extension

1. Open the Extensions panel.
2. Search for **PostgreSQL** by **Microsoft**.
3. Click **Install**.

Step 3 — Connect to the AGE database

1. Press **F1**.
2. Type **PostgreSQL: New Connection**.
3. Enter the connection details:

```
Host:      localhost
Port:      5433
User:      postgres
Password:  postgres
Database:  demo
```

You should now be connected.

4. Load and Use Apache AGE

Once connected, every session that uses AGE must load the extension and set the search path. Run the following at the start of each session:

```
-- Load the AGE extension
LOAD 'age';

-- Add AGE to the search path so graph functions are accessible
SET search_path = ag_catalog, "$user", public;
```

Create a graph

```
SELECT create_graph('my_graph');
```

Create vertices

```
SELECT * FROM cypher('my_graph', $$
    CREATE (:Person {name: 'Alice', age: 30})
$$) AS (result agtype);
```

Create edges

```
SELECT * FROM cypher('my_graph', $$
    MATCH (a:Person {name: 'Alice'})
    CREATE (a)-[:KNOWS]->(:Person {name: 'Bob', age: 25})
$$) AS (result agtype);
```

Query the graph

```
SELECT * FROM cypher('my_graph', $$  
  MATCH (a:Person)-[:KNOWS]->(b:Person)  
  RETURN a.name, b.name  
$$) AS (person1 agtype, person2 agtype);
```